IT304 LAB 10
Analysis of Ethernet frames and various network topologies.

1 To study and analyze Ethernet frames through Wireshark.

1.1 Exercise

1. First, make sure your browser's cache is empty. (To do this under Netscape 7.0, select Edit−− >Preferences− >Adv and clear the memory and disk cache. For Internet Explorer, select Tools− >Internet Options− >Delete Files. For Firefox select Tools− >Clear Private Data.
2. Start up the Wireshark packet sniffer
3. Enter the following URL into your browser http://gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html Your browser should display the rather lengthy US Bill of Rights.
4. Stop Wireshark packet capture. First, find the packet numbers (the leftmost column in the upper Wireshark window) of the HTTP GET message that was sent from your computer to gaia.cs.umass.edu, as well as the beginning of the HTTP response message sent to your computer by gaia.cs.umass.edu. You should see a screen that looks something like this (where packet 4 in the screen shot below contains the HTTP GET message)
5. Since this lab is about Ethernet and ARP, we're not interested in IP or higherlayer protocols. So let's change Wireshark's "listing of captured packets" window so that it shows information only about protocols below IP. To have Wireshark do this, select Analyze-¿Enabled Protocols. Then uncheck the IP box and select OK.
6. Select the Ethernet frame containing the HTTP GET message.
7. Expand the Ethernet II information in the packet details window. Note that the contents of the Ethernet frame (header as well as payload) are displayed in the packet contents window.

1.2 Questions

1. What is the 48-bit Ethernet address of your computer?
2. What is the 48-bit destination address in the Ethernet frame? Is this the Ethernet address of gaia.cs.umass.edu? (Hint: the answer is no). What device has this as its Ethernet address?
3. Give the hexadecimal value for the two-byte Frame type field. What do the bit(s) whose value is 1 mean within the flag field?
4. How many bytes from the very start of the Ethernet frame does the ASCII "G" in "GET" appear in the Ethernet frame?
5. What is the hexadecimal value of the CRC field in this Ethernet frame?

6. What is the value of the Ethernet source address? Is this the address of your computer, or of gaia.cs.umass.edu (Hint: the answer is no). What device has this as its Ethernet address?

7. What is the destination address in the Ethernet frame? Is this the Ethernet address of your computer?

8. Give the hexadecimal value for the two-byte Frame type field. What do the bit(s) whose value is 1 mean within the flag field?

9. How many bytes from the very start of the Ethernet frame does the ASCII "O" in "OK" (i.e., the HTTP response code) appear in the Ethernet frame?

10. What is the hexadecimal value of the CRC field in this Ethernet frame?

2 Study Star, Bus and Ring topologies and analyze their throughputs

2.1 Introduction

- **Star topology**: In local area networks with a star topology, each network host is connected to a central switch with a point-to-point connection. In Star topology every node (computer workstation or any other peripheral) is connected to central node called switch. The switch is the server and the peripherals are the clients. The network does not necessarily have to resemble a star to be classified as a star network, but all of the nodes on the network must be connected to one central device. All traffic that traverses the network passes through the central switch. The star topology is considered the easiest topology to design and implement. An advantage of the star topology is the simplicity of adding additional nodes

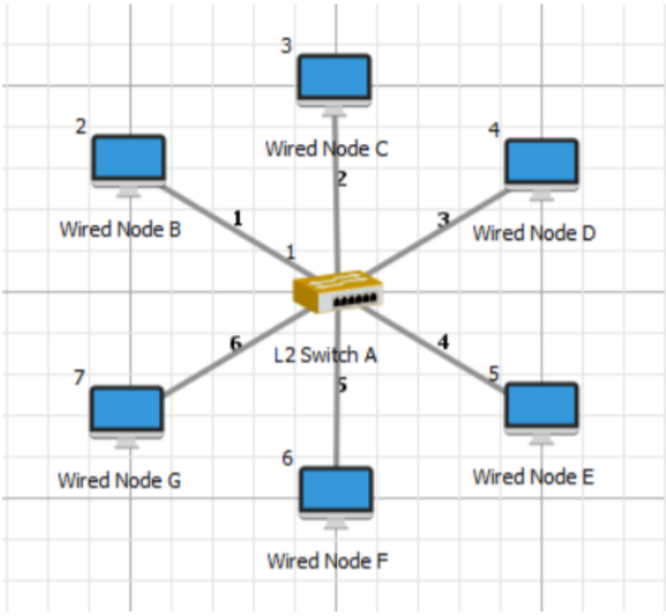
- **Token bus**:The token bus network is a standard in which tokens are passed along a virtual ring. In the token bus network bus topology is used as physical media.

  In this, the virtual ring is created with stations and therefore tokens are subsequently passed from a station during a sequence with this virtual ring. Every station or node in toke bus network knows the address of its predecessor station and its successor station. A node (station) can transmit the data if and only if it has a token. It's working rule is analogous to the token ring network.

- **Token ring:** It is defined by the IEEE 802.5 standard. In the token ring network the token is passed over a physical ring instead of virtual ring. In this a token is a special frame and a station can transmit the data frame if and only if it has a token. And The tokens are issued on successful receipt of the data frame.

2.1.1 Experiment1

1. Design a star topology
    - New → Internetworks
    - Implement following topology



2. Switch Properties: Accept the default properties of switch
3. Wirednode properties: Disable TCP at transport layer in all wired nodes
4. Link properties: Uplink and downlink speed=100 Mbps, BER=no error

5. Application properties:

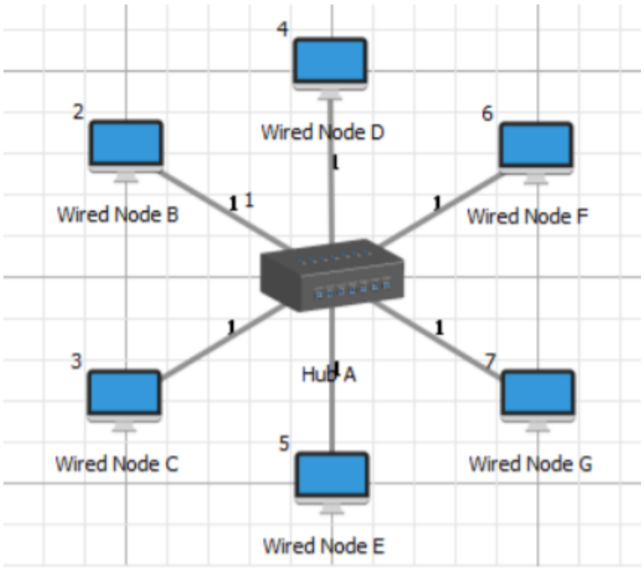
Application Properties:

Application Properties	2(Wired Node B)	3(Wired Node C)	4(Wired Node D)	5(Wired Node E)	6(Wired Node F)	7(Wired Node F)
Destination	3(Wired Node C)	4(Wired Node D)	5(Wired Node E)	6(Wired Node F)	7(Wired Node B)	2(Wired Node B)
Application Type	Custom	Custom	Custom	Custom	Custom	Custom
Packet Size						
Distribution	Constant	Constant	Constant	Constant	Constant	Constant
Packet Size (Bytes)	10000	10000	10000	10000	10000	10000
Packet Inter Arrival Time						
Distribution	Constant	Constant	Constant	Constant	Constant	Constant
Packet Inter Arrival Time (μs)	1000	1000	1000	1000	1000	1000

6. Simulation time=10s
7. Note down average throughput of all application

2.1.2 Experiment2

1. New→ Legacy Network →Token Bus
2. Implement following topology

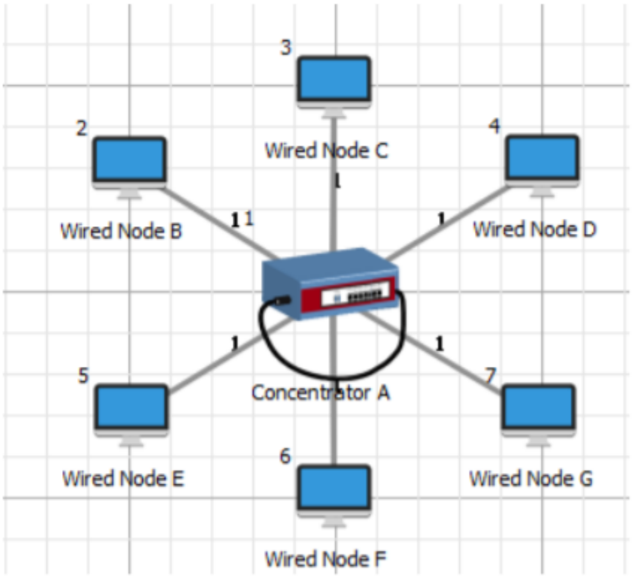


3. Wirednode properties: Disable TCP at transport layer in all wired nodes
4. Link properties: Uplink and downlink speed=10 Mbps, BER=no error
5. Application properties:Same as previous experiment

6. Simulation time=10s
7. Note down average throughput of all application

2.1.3 Experiment 3

1. Create scenario: New → Legacy network → Token ring
2. Implement following topology:



3. Concentrator properties(set by link properties):

Concentrator Properties:

Concentrator Properties	Values to be Selected
Data Rate(Mbps)	16
Error Rate (bit error rate)	No error

Figure 1: Caption

4. Link properties: Uplink and downlink speed=100 Mbps, BER=no error
5. Application properties:As previous experiment
6. Simulation time=10s
7. Note down average throughput of all application

Compare Throughtput of all these three experiment and justify the output.

3 Submission guidelines

Submit a pdf file which contains answer of all exercises.